ISTE-140 WEB & MOBILE I · HOMEWORK 05

HW 05: Working with Linux/Unix

Terminal vs. FileZilla — one server, two views

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Your first day at Tiger Web Studio

Kavana Bura, a (fictional) café in Zagreb, has hired Tiger Web Studio to build its first website — and you are the studio's newest junior developer. Before anyone writes a line of CSS, the project has to exist on the server: folders created, files in place, a backup made, a secret file locked down and permissions set so the site could go live.

You will do all of it from the command line on **solace**, and you will double-check every step in **FileZilla**, a graphical client connected to the very same server. By the end you will know what the GUI is really doing behind the scenes — and when the terminal is simply faster.

By the end of this homework you will be able to:

- move around the Linux file system with absolute and relative paths;
- create, copy, move, rename and delete files and directories — including whole trees;
- read and set permissions in numeric and symbolic form, and explain what they protect;
- match every terminal action to what you see in FileZilla, and explain where the two differ.

Screen recording required — Part H

Record **one continuous run-through of the whole homework** with your terminal and FileZilla side by side, your webcam on and your voice explaining what you do and why.

No recording, no mark: a submission without an acceptable recording receives 0 points, however good the written answers are.

What you submit

- This Word document with every answer box filled in and the three screenshots pasted, saved as `ISTE140_HW05_Lastname_Firstname.docx`.
- Your screen recording (Part H) as an MP4 file — or, if it is too large for the dropbox, a link your instructor can open.
- Both go to the assignment dropbox before the due date.

Before you start

- **An SSH client** — the built-in `ssh` in Windows cmd / PowerShell / Windows Terminal, PuTTY, or Terminal on macOS and Linux (see LN05, slides 7–8).
- **FileZilla Client** (free) from filezilla-project.org. Install the *Client*, not the Server, and decline any bundled extra software the installer offers.
- Your **RIT username and password** — the same ones you use for `ssh`.

- **A screen recorder** that captures the whole screen with microphone — OBS Studio (free, Windows/macOS/Linux) can also overlay your webcam. Windows Snipping Tool or macOS Cmd+Shift+5 also work; if your recorder cannot overlay the webcam, open your camera app in a small window on screen.

Allowed commands — only what we used in LN05 Commands and Files

Combine options as in class (for example `ls -la`, `ls -ld`). A step solved with any other command or tool — for example nano, vi, echo, find, tree or sudo — earns no points. In FileZilla you may use any menu item.

Command	Use it to	Command	Use it to
`ssh`	connect to solace	`pwd`	print the working directory
`cd`	change directory: `..` one level up, `/path` absolute, nothing = home	`ls`	list: `-a` all, `-l` long, `-d` the directory itself, `-h` human sizes, `-s` size, `-S` sort by size, `-t` sort by time
`touch`	create an empty file or update its time	`mkdir`	make a directory; `-p` creates missing parents
`rmdir`	remove an empty directory; `-p` also its parents	`mv`	rename or move
`cp`	copy; `-r` copies a whole directory	`rm`	delete; `-r` deletes a whole directory
`cat`	print the contents of one or more files	`chmod`	change permissions: numeric (`755`) or symbolic (`u+x`)
`man`	read the manual of a command (`q` quits)		

Golden rule: there is no recycle bin on the server

Do all your work inside **hw05**. Before every `rm` or `rm -r`, run `pwd` and `ls` to make sure you are exactly where you think you are. Do not touch the files you published in HW04.

Academic integrity and AI use

This is an individual assignment. You may use the `man` pages, your classmates or an AI assistant to understand a concept or an error message. You may not copy someone else's answers or paste commands you do not understand. Your written answers must match what you do and say in your recording — your username is visible in the prompt, and your voice explains why each command works. Parts where the recording and the written answers do not match earn no points.

PART A Connect and get your bearings

8 pts

A1 Open your SSH client and log in with `ssh yourusername@solace.ist.rit.edu`. While you type the password nothing appears — no dots, no stars — that is normal. Then run the command that prints your current working directory. (2 pts)

Command and its exact output

```
[ssk6197@ist-solace ~]$ pwd
/home/MAIN/ssk6197
[ssk6197@ist-solace ~]$
```

Your prompt ends with `~` (e.g. `[abc1234@ist-solace ~]$`). What does the `~` stand for? Compare it with the output above.

Your answer

~ means that we are in our home directory comparing with the output above. we can see that our current working directory is our home directory

A2 Start FileZilla and fill in the **Quickconnect** bar at the top: Host `sftp://solace.ist.rit.edu`, Username = your RIT username, Password = your password, Port `22`. Click **Quickconnect**. If FileZilla asks whether to trust an unknown host key, check that the host is solace.ist.rit.edu and confirm. The server appears in the right-hand **Remote site** pane. (2 pts)

What path is shown in the **Remote site** box? Is it identical to your answer in A1? Why does it make sense that both tools start in the same place?

Your answer

/home/MAIN/ssk6197. it is identical to A1. They start at the same place because because it is our home directory

> **Behind the scenes**
>
> FileZilla's `sftp://` stands for the **SSH File Transfer Protocol**. It uses the same port (22), the same account and the same password as your `ssh` session and lands in the same home directory. One server — two windows into it. Every click in FileZilla becomes a request to the server, just like the commands you type.

A3 In your home directory run `ls`, then `ls -a`, then `ls -la`. (2 pts)

(a) Which names appear only with `-a`, and what do they all have in common? (b) Look at FileZilla's Remote site file list. Which of the three commands lists the same entries as FileZilla?

(a) and (b)

(a)ls -a shows us files starting with . , these are hidden files not normally visible when we enter a directory. filezilla lists all the files from ls -la command

A4 Take the tour from class: `cd /usr/lib`, `pwd`, `cd ..`, `pwd`, then `cd` with nothing after it and `pwd` once more. Now run `ls -ld /usr/lib`. (2 pts)

(a) Who owns /usr/lib, and which permissions does the *others* category (that is you) have? (b) Predict whether you can create a directory there, then test it with `mkdir /usr/lib/hw05test` and copy the exact message. (c) In FileZilla, type /usr/lib into the Remote site box and press Enter, then try right-click → **Create directory**. What happens, and where does FileZilla report it?

(a), (b) and (c)

I do not own it. My permissions (others) is read and execute only.

(b) mkdir: cannot create directory '/usr/lib/hw05test': Permission denied

(c) permission denied, filezilla reports at the status box.
Could not create directory: Received error SSH_FX_PERMISSION_DENIED with description 'Permission denied'

PART B Build the client project

12 pts

Build this structure inside your home directory. Names ending with `/` are directories; everything else is an empty file for now.

```
hw05/
├── drafts/
└── kavana-bura/
    ├── index.html
    ├── menu.html
    ├── contact.html
    ├── README.txt
    ├── css/
    │   └── style.css
    ├── js/
    │   └── app.js
    └── assets/
        └── img/
            └── menu/
```

B1 Go back to your home directory with a single command, create `hw05` and make it your working directory. Confirm with `pwd`. (2 pts)

Commands

```
[ssk6197@ist-solace ~]$ cd
[ssk6197@ist-solace ~]$ mkdir hw05
[ssk6197@ist-solace ~]$ cd hw05/
[ssk6197@ist-solace hw05]$ pwd
/home/MAIN/ssk6197/hw05
[ssk6197@ist-solace hw05]$
```

B2 Create every directory in the tree. Rules: use `mkdir -p` at least once to create a nested path in one command, and do not `cd` into a directory just to create something inside it — use paths instead. (4 pts)

Command golf: write your commands one per line and count them. Fewest commands wins (bragging rights only).

Commands (one per line) and total count

```
mkdir drafts
mkdir kavana-bura
mkdir kavana-bura/css
mkdir kavana-bura/js
mkdir kavana-bura/assets
mkdir kavana-bura/assets/img
mkdir kavana-bura/assets/img/menu
```

B3 Create all six empty files with `touch`. One `touch` can take several names, and a name can include a path (for example `css/style.css`). (2 pts)

Commands

```
touch index.html menu.html contact.html README.txt css/style.css js/app.js
```

B4 Inside kavana-bura run `ls -l`, then `ls -d */`. (2 pts)

(a) What does `ls -d */` show, and how is that different from `ls -l`? (b) In the `ls -l` output, how do you tell a directory from a regular file? (c) What size does `ls -l` report for your new files, and why?

(a), (b) and (c)

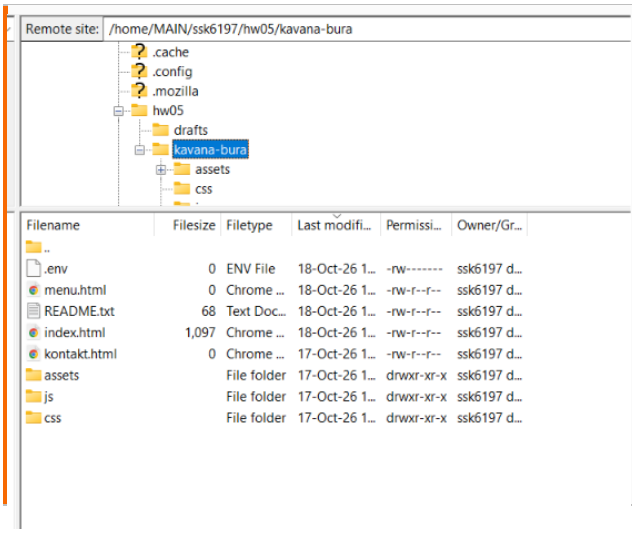
(a) it shows only the directories, ls -l shows a detailed list of files including permissions
(b) directories have / at the end and files have a . extension like .txt or .html
(c)the size for the new files are 0 as there is nothing in it.

B5 Switch to FileZilla. Is hw05 already in the Remote site list? Press **F5** (Refresh) and look again. Then open kavana-bura in FileZilla. (2 pts)

Which FileZilla column matches each part of an `ls -l` line: permissions, owner and group, size, date/time, name? Did you need F5 before hw05 appeared?

Your answer

filezilla col 5- permissions
filezilla col 6- owner/group
filezilla col 2 - size
col 4 - date/time
col 1 - name



FileZilla showing the same folder

PART C Navigate like a pro: relative vs. absolute paths 10 pts

This is the table from LN05 slide 6 — now with your own project. Before each row, `cd` to the **Current location** (any way you like). Then write the **relative** and the **absolute** `cd` command that take you to the **Desired location** and verify each with `pwd`. Locations are shown relative to hw05; absolute paths start with `/` and use your home path from A1.

C1 Complete the table (row 6: you also name the relationship). (8 pts)

#	Current location	Desired location	Relationship	Relative path	Absolute path
1	kavana-bura/css	kavana-bura	Parent	cd ..	cd /home/MAIN/ssk6197/hw05/kavana-bura
2	kavana-bura	kavana-bura/assets/img/menu	Great-grandchild	cd assets/img/menu/	cd /home/MAIN/ssk6197/hw05/kavana-bura/assets/img/menu
3	kavana-bura/css	kavana-bura/js	Sibling	cd ../js/	cd /home/MAIN/ssk6197/hw05/kavana-bura/js
4	kavana-bura/assets/img/menu	kavana-bura	Great-grandparent	cd ../../..	cd /home/MAIN/ssk6197/hw05/kavana-bura
5	kavana-bura/js	drafts	Parent's sibling	cd ../../drafts/	cd /home/MAIN/ssk6197/hw05/drafts

#	Current location	Desired location	Relationship	Relative path	Absolute path
6	kavana-bura/assets/img/menu	kavana-bura/js	?	cd ../../../js/	cd /home/MAIN/ssk6197/hw05/kavana-bura/js
7	anywhere	your home directory	—	cd	cd /home/MAIN/ssk6197

C2 Give one situation where an absolute path is the better choice and one where a relative path is better. (2 pts)

Your answer

if the path is too long to use relative path. absolute path is easier as we know the location does not change

PART D Client changes: rename, move, copy, delete 14 pts

The client has sent a list of changes. Work from inside **hw05** unless a step says otherwise.

D1 The client wants Croatian URLs. Rename `contact.html` to `kontakt.html`. (1 pt)

Command

mv contact.html kontakt.html

D2 A teammate created a `fonts` directory in the wrong place. Simulate the mistake: from hw05 run `mkdir fonts`. It belongs in kavana-bura/assets. Fix it with **one** command, then prove it with `ls`. (2 pts)

Commands and ls output

mv fonts/ kavana-bura/assets/

D3 Before the next round of changes, make a backup: copy the whole kavana-bura directory, with everything inside it, to `kavana-bura-backup`. Verify with `ls`. (1 pt)

cp -r kavana-bura/ kavana-bura-backup

D4 **The double-backup trap.** Run exactly the same command as in D3 a second time, then run `ls kavana-bura-backup`. (3 pts)

(a) What is new inside the backup? (b) Why did `cp` behave differently the second time? (c) Remove *only* the accidental extra copy with a single command — run `pwd` and `ls` first.

(a)the entire kavana-bura directory got copied inside kavana-bura-backup.
(b)the second time did not create a new directory, but copied the directory into the existing one
(c)rm -r kavana-bura-backup/kavana-bura/

D5 Copy `kavana-bura/menu.html` into drafts under the new name `menu-v2.html` — one command, run from hw05. (1 pt)

cp kavana-bura/menu.html drafts/menu-v2.html

D6 **The rmdir puzzle.** Create `drafts/archive/2025/q4` with one command. Then run `rmdir -p drafts/archive/2025/q4` and read the message carefully. (3 pts)

(a) Which directories were removed? (b) Which one survived, and why? (c) Which LN05 command would remove drafts even though it is not empty — and why should you not run it now?

(a)none of them were removed. rmdir: failed to remove directory 'drafts': Directory not empty
(b)they all survived, only empty directories can be removed from this command
(c) rm -r removes the directory. the change is not reversible

D7 Now the same kind of work in FileZilla. (3 pts)

(a) Try to make a copy of `kavana-bura/index.html` on the server using only FileZilla — no downloading and re-uploading. Could you? Which terminal command does it in one step?

(b) Delete kavana-bura-backup in FileZilla (right-click → **Delete**). What did FileZilla ask you? Which single terminal command does the same thing, and did the terminal ask you anything when you used it in D4? Confirm with `ls` in the terminal that the backup is gone.

(a)It does not work. you cannot copy a file without downloading on filezilla. in ssh cp index.html index2.html in one command
(b)it asked for confirmation and deleted the dir. rm -p kavana-bura-backup does the same thing

PART E Content, sizes and timestamps — 10 pts

Your files are still empty. The commands from LN05 cannot type text into a file — but FileZilla can bring content in from your own computer. Watch how the terminal sees every change.

E1 On your own computer, create `index.html` in your editor: a minimal page (you can reuse your HW02 boilerplate) with the title **Kavana Bura — coming soon** and your full name in an `<h1>`. In FileZilla, drag it from the Local site pane into kavana-bura on the Remote site and choose **Overwrite**. In the terminal run `ls -l index.html` and `cat index.html`. (2 pts)

What was the file size before (B4) and after the upload? Does `cat` show your page?

Yes, cat shows my page. the file size changed from 0 to 1097 bytes

E2 In FileZilla right-click `README.txt` → **View/Edit** (if asked which program to use, choose a text editor such as Notepad, VS Code or TextEdit). Type three lines: the client's name, your name and today's

date. Save, switch back to FileZilla and confirm that the changed file should be uploaded back to the server. In the terminal run `cat README.txt`, then `cat README.txt index.html`. (2 pts)

What does `cat` do when you give it two file names?

The cat command concatenates two files and prints on screen.

E3 Run `ls -l menu.html`, then `touch menu.html` (the file already exists), then `ls -l menu.html` again. (2 pts)

What changed and what stayed the same? Does FileZilla's **Last modified** column update by itself, or only after F5?

(a)the date modified has changed. Filezilla only updates after pressing f5.

E4 In kavana-bura run `ls -lS`, `ls -lt` and `ls -lh`. Then click FileZilla's **Filesize** and **Last modified** column headers. (2 pts)

Which file is the biggest and which is the newest? Does FileZilla sort them in the same order as `ls -S` and `ls -t`?

The biggest file is index.html and the newest is menu.html. Yes, filezilla sort them in the same order as ls -S and ls -t

E5 **Detective question.** Run `ls -s index.html` and `ls -l index.html`. The numbers differ. Use `man ls` (press `q` to quit) to find out what `-s` measures and explain the difference in one sentence. (2 pts)

-s gives us the size of the file in disk blocks. it shows us how much it takes up space on the disk instead of bytes.

PART F Permissions and security

16 pts

F1 Read the defaults with `ls -ld kavana-bura` and `ls -l kavana-bura`, and complete the table. (2 pts)

Item	Permission string (e.g. drwx------)	Numeric	Created with
kavana-bura (directory)	drwxr-xr-x	4	`mkdir`
menu.html	-rw-r--r--	1	`touch`
index.html	-rw-r--r--	1	FileZilla upload

Are menu.html and index.html the same? If not, suggest why.

No, they are not the same. The contents of the files and the file size are different. the only thing that is similar is that they are both html files.

F2 **The secret file.** Real projects keep passwords and API keys in a file called `.env`. Create an empty `.env` in kavana-bura. Compare `ls` and `ls -a` — and check whether FileZilla shows it. Then lock it so that only you can read and write it, using numeric mode. (2 pts)

Write the `chmod` command and the resulting permission string. Then explain in one sentence why *hidden* is not the same as *protected*.

chmod 00600. Hidden files do not mean that they are protected as using ls-a can reveal any hidden files. changing the permissions protects the file

F3 **Go-live ready.** A web server runs as a different user than you, so it falls into the *others* category. Using **numeric** mode, set every directory inside kavana-bura (including kavana-bura itself, and the fonts directory you moved in D2) to `rwxr-xr-x`, and every .html, .css, .js and .txt file to `rw-r--r--`. Leave `.env` as it is. (3 pts)

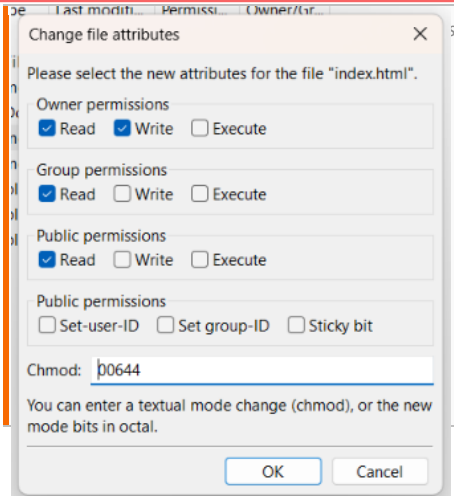
Write your commands. Then explain why *others* need **read** on files but **execute** on directories for a page to load.

chmod 755 for dir
chmod 644 for files
other need only read on files as they do not need to write on them and mess with the files.execute is needed on directories to let them open them

F4 In FileZilla right-click `index.html` → **File permissions…**. Now use the same dialog on `menu.html` to remove **Read** for Group and Public, and click OK. (2 pts)

(a) Which numeric value does FileZilla show for index.html? (b) What does `ls -l menu.html` show now? (c) Restore menu.html to `rw-r--r--` with a single **symbolic** `chmod` (no numbers), then press F5 in FileZilla to confirm.

(a)644 (b)-rw-------. 1 ssk6197 domain^users 0 Oct 18 05:36 menu.html



ssions dialog for index.html

F5 **Predict, then verify.** Scripts do not belong in a public site folder, so in hw05 (not in kavana-bura) run `touch deploy.sh`. For each row, write your **prediction first**, then run the command and copy what `ls -l deploy.sh` really shows. (3 pts)

Command (run in order)	Predicted string	Actual string (ls -l)	Numeric
`chmod 600 deploy.sh`	-rw-------	-rw-------	
`chmod u+x deploy.sh`	-rwx------	-rwx------	
`chmod go=r deploy.sh`	-rw----r--.	-rwxr--r--	
`chmod a-w deploy.sh`	-r-xr--r--	-r-xr--r--	
`chmod ug+w deploy.sh`	-rwxrw-r--	-rwxrw-r--	

F6 **Locked out.** Run `chmod 000 drafts`. Then try `ls drafts`, `cd drafts`, and opening drafts in FileZilla. Finally restore it with `chmod 700 drafts`. (2 pts)

Copy the three error messages (FileZilla shows its error in the message log at the top). You had no permissions at all on drafts — so why were you still allowed to run `chmod` on it?

ls: cannot open directory drafts/: Permission denied
-bash: cd: drafts/: Permission denied
Could not open directory: Received error SSH_FX_PERMISSION_DENIED with description 'Permission denied'
i can change chmod because i still had access to the remote server

F7 **The delete puzzle.** Run `touch drafts/trash.txt` and `chmod 444 drafts/trash.txt`, then try `rm drafts/trash.txt` — `rm` asks a question: answer `n`. Now run `chmod 500 drafts` and try the same `rm` again (answer `y` if asked). Finally run `chmod 700 drafts` and try once more (answer `y`). (2 pts)

What happened each time? Which permission really decides whether a file can be deleted — the file's own or its directory's? Compare with the Permissions Interpretation table in LN05 (slide 13).

The first time chmod 444 gives read to all ugo. but the second chmod 500 gives user r and x. as we could not write on it we could not delete the file. and after changing to 700 we could delete it. the directories permission decided if the file can be deleted or not

PART G Final audit: terminal vs. FileZilla

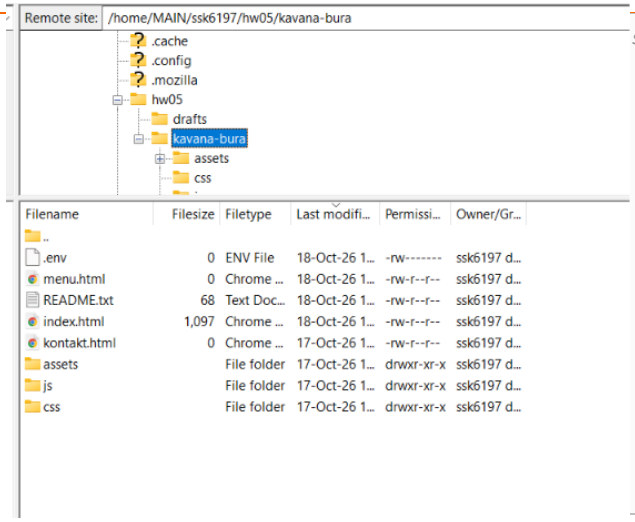
10 pts

G1 Fill in the comparison table from your own experience in this homework. (5 pts)

Task	Terminal command	FileZilla action	Faster or clearer? Why?
Show where I am on the server	pwd	remote site:	pwd. it cal tell in one command
List all files, including hidden ones	ls -la	open the directory	filezilla. it shows hidden files by
Create a nested path (a/b/c) in one step	mkdir a mkdir a/b mkdir a/b/c	just open the dir and make a new one. once you reach c make a file	filezilla. we can just create dir easily and open and create a file
Create an empty file	cd a/b/c touch file		
Rename a file	mv file file2	right click rename	both are fast
Move a directory into another directory	mv dir1 dir2	drag and drop or cut and paste	filezilla is faster
Copy a file on the server	cp file1 file2	right click copy	both are fast
Delete a directory and everything in it	rmdir -p dir	right click delete	both are fast
Read the contents of a text file			
Set a file's permissions to 644	chmod 644 file	right click file permissions and change	terminal is faster but filezilla is
Find the biggest and the newest file	ls -S ls -t	click on the column	filezilla is faster as it is a single click

G2 From hw05 run `ls -la kavana-bura` and paste the text output below. Then refresh FileZilla (F5) on the same folder and take the final screenshot. (2 pts)

```
total 8
drwxr-xr-x. 5 ssk6197 domain^users  126 Oct 18 05:54 .
drwxr-xr-x. 4 ssk6197 domain^users   56 Oct 18 06:14 ..
drwxr-xr-x. 4 ssk6197 domain^users   30 Oct 17 17:49 assets
drwxr-xr-x. 2 ssk6197 domain^users   23 Oct 17 17:11 css
-rw-------. 1 ssk6197 domain^users    0 Oct 18 05:54 .env
-rw-r--r--. 1 ssk6197 domain^users 1097 Oct 18 05:26 index.html
drwxr-xr-x. 2 ssk6197 domain^users   20 Oct 17 17:11 js
-rw-r--r--. 1 ssk6197 domain^users    0 Oct 17 17:11 kontakt.html
-rw-r--r--. 1 ssk6197 domain^users    0 Oct 18 05:36 menu.html
-rw-r--r--. 1 ssk6197 domain^users   68 Oct 18 05:32 README.txt
```



side, same folder

G3 Reflection — two or three sentences each. (3 pts)

(a) Name one moment in this homework when FileZilla showed you something out of date. How did you fix it, and why does it happen?

i did not encounter anything out of date this homework

(b) Name one real situation in a web developer's job where you will *not* have FileZilla — only a terminal.

I cannot think of any situation where you would have only one and not the other

(c) Which command in this homework is the most dangerous, and which habit protects you from it?

rm command might be the most dangerous and making backups will protect us from accidentally deleting the files

PART H Screen recording — show the whole homework 20 pts

Record **one continuous run-through of Parts A–G**, from login to the final side-by-side, narrated in your own words. Do the homework once to fill in this document, then record yourself doing it again as your clean, final run. A good recording sounds like you are teaching a classmate: say what you are about to do and why — do not type in silence.

Set up before you press record

- Connect FileZilla **before** recording so your password is never on screen. Place the terminal and FileZilla side by side.
- Make the terminal readable: enlarge its font (Ctrl + mouse wheel in Windows Terminal, or the settings in PuTTY/Terminal).
- Webcam on in a corner of the screen, microphone on, notifications off, all other windows closed.
- Start from a clean slate: if hw05 still exists from your practice run, the recording begins with `pwd`, `ls`, `rm -r hw05` and `ls` again to show it is gone.

Must-show checkpoints

Part	What the viewer must see and hear
A	SSH login, `pwd` and `ls -la`; FileZilla showing the same home path; explain `~`.
B	The whole tree built from scratch; say out loud what `-p` does. F5 in FileZilla.
C	At least three rows of the navigation table, each confirmed with `pwd`.
D	Rename and move; the double-backup trap and its fix; the rmdir puzzle; deleting the backup in FileZilla and confirming it in the terminal.
E	Upload index.html, edit README.txt with View/Edit, then `cat` both in the terminal; `touch` and the timestamp change.
F	The `.env` lock-down, the web-ready permissions, FileZilla's File permissions dialog, the locked-out experiment and the delete puzzle — explain each result.
G	Final `ls -la` next to the refreshed FileZilla window. End with "HW05 complete".

Recording rules

- One continuous take — no cuts, no edits, no speed-up. **Maximum 15 minutes.** Practise first.
- Your username must be visible in the terminal prompt and your face in the webcam overlay for the whole recording.
- MP4, 1080p if possible (720p is acceptable), named `ISTE140_HW05_Lastname_Firstname.mp4`.
- If the file is too large for the dropbox, upload it to OneDrive or Google Drive, give your instructor view access and paste the link below.

> *Recording link (only if you did not upload the MP4 file)*

HW 05: Working with Linux/Unix — Grading

Name: _____ *RIT username:* _____

Part	What is assessed	Points	Score
A	A1–A4: login and `pwd`, `~` explained, FileZilla connected to the same home, hidden files, reading /usr/lib permissions	8	
B	B1–B5: hw05 created, full tree with `mkdir -p`, files with `touch`, `ls -d */` and `ls -l` explained, column mapping, Screenshot 1	12	
C	C1 navigation table (7 rows, relative and absolute, verified with `pwd`); C2 absolute vs. relative	10	
D	D1–D7: rename, move, `cp -r` backup, double-backup trap fixed precisely, copy with new name, rmdir puzzle explained, FileZilla copy/delete compared	14	
E	E1–E5: upload and `cat`, View/Edit round trip, `touch` and timestamps, sorting by size and time, `ls -s` vs. `ls -l`	10	
F	F1–F7: default permissions, `.env` locked down, web-ready 755/644, FileZilla dialog and Screenshot 2, predict-and-verify table, locked-out experiment, delete puzzle	16	
G	G1 comparison table, G2 final listing and Screenshot 3, G3 reflection	10	
H	Recording: complete run-through of A–G (8) · clear explanations of *why* at the checkpoints (6) · rules met: one take, ≤ 15 min, readable terminal, webcam and voice (4) · final state matches this document (2)	20	
	Total	**100**	

No recording, no mark

Without an acceptable recording (Part H), the whole homework receives **0 points**, however complete the written answers are.

Steps solved with commands outside the allowed LN05 list earn no points, and written answers that the recording contradicts earn no points.